Solution By the definition of division we obtain

$$\frac{1}{z} = \frac{1}{2-3i} = \frac{1}{2-3i}\,\frac{2+3i}{2+3i} = \frac{2+3i}{4+9} = \frac{2+3i}{13}.$$

Answer should be in the form $a+ib$. ▶ That is,

$$\frac{1}{z} = z^{-1} = \frac{2}{13} + \frac{3}{13}i.$$

You should take a few seconds to verify the multiplication

$$zz^{-1} = (2-3i)\left(\tfrac{2}{13} + \tfrac{3}{13}i\right) = 1.$$

☰

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) Many of the properties of the real number system **R** hold in the complex number system **C**, but there are some truly remarkable differences as well. For example, the concept of *order* in the real number system does not carry over to the complex number system. In other words, we cannot compare two complex numbers $z_1 = a_1 + ib_1$, $b_1 \neq 0$, and $z_2 = a_2 + ib_2$, $b_2 \neq 0$, by means of inequalities. Statements such as $z_1 < z_2$ or $z_2 \geq z_1$ have no meaning in **C** except in the special case when the two numbers z_1 and z_2 are real. See Problem 55 in Exercises 1.1. Therefore, if you see a statement such as $z_1 = \alpha z_2$, $\alpha > 0$, it is implicit from the use of the inequality $\alpha > 0$ that the symbol α represents a real number.

(*ii*) Some things that we take for granted as impossible in real analysis, such as $e^x = -2$ and $\sin x = 5$ when x is a real variable, are perfectly correct and ordinary in complex analysis when the symbol x is interpreted as a complex variable. See Example 3 in Section 4.1 and Example 2 in Section 4.3.

We will continue to point out other differences between real analysis and complex analysis throughout the remainder of the text.

EXERCISES 1.1

Answers to selected odd-numbered problems begin on page ANS-1.

1. Evaluate the following powers of i.

(a) i^8 **(b)** i^{11}

(c) i^{42} **(d)** i^{105}

2. Write the given number in the form $a+ib$.

(a) $2i^3 - 3i^2 + 5i$ **(b)** $3i^5 - i^4 + 7i^3 - 10i^2 - 9$

(c) $\dfrac{5}{i} + \dfrac{2}{i^3} - \dfrac{20}{i^{18}}$ **(d)** $2i^6 + \left(\dfrac{2}{-i}\right)^3 + 5i^{-5} - 12i$

In Problems 3–20, write the given number in the form $a + ib$.

3. $(5 - 9i) + (2 - 4i)$

4. $3(4 - i) - 3(5 + 2i)$

5. $i(5 + 7i)$

6. $i(4 - i) + 4i(1 + 2i)$

7. $(2 - 3i)(4 + i)$

8. $\left(\frac{1}{2} - \frac{1}{4}i\right)\left(\frac{2}{3} + \frac{5}{3}i\right)$

9. $3i + \dfrac{1}{2 - i}$

10. $\dfrac{i}{1 + i}$

11. $\dfrac{2 - 4i}{3 + 5i}$

12. $\dfrac{10 - 5i}{6 + 2i}$

13. $\dfrac{(3 - i)(2 + 3i)}{1 + i}$

14. $\dfrac{(1 + i)(1 - 2i)}{(2 + i)(4 - 3i)}$

15. $\dfrac{(5 - 4i) - (3 + 7i)}{(4 + 2i) + (2 - 3i)}$

16. $\dfrac{(4 + 5i) + 2i^3}{(2 + i)^2}$

17. $i(1 - i)(2 - i)(2 + 6i)$

18. $(1 + i)^2(1 - i)^3$

19. $(3 + 6i) + (4 - i)(3 + 5i) + \dfrac{1}{2 - i}$

20. $(2 + 3i)\left(\dfrac{2 - i}{1 + 2i}\right)^2$

In Problems 21–24, use the **binomial theorem***

$$(A + B)^n = A^n + \frac{n}{1!}A^{n-1}B + \frac{n(n-1)}{2!}A^{n-2}B^2 + \cdots + \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}A^{n-k}B^k + \cdots + B^n,$$

where $n = 1,\ 2,\ 3,\ \ldots$, to write the given number in the form $a + ib$.

21. $(2 + 3i)^2$

22. $\left(1 - \frac{1}{2}i\right)^3$

23. $(-2 + 2i)^5$

24. $(1 + i)^8$

In Problems 25 and 26, find $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$.

25. $z = \left(\dfrac{i}{3 - i}\right)\left(\dfrac{1}{2 + 3i}\right)$

26. $z = \dfrac{1}{(1 + i)(1 - 2i)(1 + 3i)}$

In Problems 27–30, let $z = x + iy$. Express the given quantity in terms of x and y.

27. $\mathrm{Re}(1/z)$

28. $\mathrm{Re}(z^2)$

29. $\mathrm{Im}(2z + 4\bar{z} - 4i)$

30. $\mathrm{Im}(\bar{z}^2 + z^2)$

In Problems 31–34, let $z = x + iy$. Express the given quantity in terms of the symbols $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$.

31. $\mathrm{Re}(iz)$

32. $\mathrm{Im}(iz)$

33. $\mathrm{Im}((1 + i)z)$

34. $\mathrm{Re}(z^2)$

In Problems 35 and 36, show that the indicated numbers satisfy the given equation. In each case explain why additional solutions can be found.

35. $z^2 + i = 0$, $z_1 = -\dfrac{\sqrt{2}}{2} + \dfrac{\sqrt{2}}{2}i$. Find an additional solution, z_2.

36. $z^4 = -4$; $z_1 = 1 + i,\ z_2 = -1 + i$. Find two additional solutions, z_3 and z_4.

---

*Recall that the coefficients in the expansions of $(A + B)^2$, $(A + B)^3$, and so on, can also be obtained using Pascal's triangle.

In Problems 37–42, use Definition 1.1.2 to solve each equation for $z = a + ib$.

37. $2z = i(2 + 9i)$

38. $z - 2\bar{z} + 7 - 6i = 0$

39. $z^2 = i$

40. $\bar{z}^2 = 4z$

41. $z + 2\bar{z} = \dfrac{2 - i}{1 + 3i}$

42. $\dfrac{z}{1 + \bar{z}} = 3 + 4i$

In Problems 43 and 44, solve the given system of equations for z_1 and z_2.

43.
$$\begin{aligned} iz_1 - \quad & iz_2 = 2 + 10i \\ -z_1 + (1 - i)z_2 &= 3 - 5i \end{aligned}$$

44.
$$\begin{aligned} iz_1 + (1 + i)z_2 &= 1 + 2i \\ (2 - i)z_1 + \quad & 2iz_2 = 4i \end{aligned}$$

Focus on Concepts

45. What can be said about the complex number z if $z = \bar{z}$? If $(z)^2 = (\bar{z})^2$?

46. Think of an alternative solution to Problem 24. Then without doing any significant work, evaluate $(1 + i)^{5404}$.

47. For n a nonnegative integer, i^n can be one of four values: 1, i, -1, and $-i$. In each of the following four cases, express the integer exponent n in terms of the symbol k, where $k = 0, 1, 2, \ldots$.

(**a**) $i^n = 1$

(**b**) $i^n = i$

(**c**) $i^n = -1$

(**d**) $i^n = -i$

48. There is an alternative to the procedure given in (7). For example, the quotient $(5 + 6i)/(1 + i)$ must be expressible in the form $a + ib$:

$$\frac{5 + 6i}{1 + i} = a + ib.$$

Therefore, $5 + 6i = (1 + i)(a + ib)$. Use this last result and Definition 1.1.2 to find the given quotient. Use this method to find the reciprocal of $3 - 4i$.

49. Assume for the moment that $\sqrt{1 + i}$ makes sense in the complex number system. How would you then demonstrate the validity of the equality

$$\sqrt{1 + i} = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{2}} + i\sqrt{-\tfrac{1}{2} + \tfrac{1}{2}\sqrt{2}}?$$

50. Suppose that z_1 and z_2 are complex numbers. Use Definition 1.1.2 to prove that if $z_1 z_2 = 0$, then $z_1 = 0$ or $z_2 = 0$.

51. Suppose the product $z_1 z_2$ of two complex numbers is a nonzero real constant. Show that $z_2 = k\bar{z}_1$, where k is a real number.

52. Without doing any significant work, explain why it follows immediately from (2) and (3) that $z_1\bar{z}_2 + \bar{z}_1 z_2 = 2\text{Re}(z_1\bar{z}_2)$.

53. Mathematicians like to prove that certain "things" within a mathematical system are unique. For example, a proof of a proposition such as "The unity in the complex number system is unique" usually starts out with the assumption that there exist two *different* unities, say, 1_1 and 1_2, and then proceeds to show that this assumption leads to some contradiction. Give one contradiction if it is assumed that two different unities exist.

54. Follow the procedure outlined in Problem 53 to prove the proposition "The zero in the complex number system is unique."

55. A number system is said to be an **ordered system** provided it contains a subset P with the following two properties:

First, for any nonzero number x in the system, either x or $-x$ is (but not both) in P.

Second, if x and y are numbers in P, then both xy and $x + y$ are in P.

In the real number system the set P is the set of *positive* numbers. In the real number system we say x is greater than y, written $x > y$, if and only if $x - y$ is in P. Discuss why the complex number system has no such subset P. [*Hint*: Consider i and $-i$.]

Computer Lab Assignments

In Problems 56–59, use a CAS* to write the given complex number in the form $a + bi$. In *Mathematica* use the command **ComplexExpand**.

56. $2 + i - \dfrac{1}{3 - i}$ **57.** $\dfrac{5i}{2 + 3i}$

58. $\dfrac{1 + i}{2 - 5i}$ **59.** $\dfrac{(1 + 3i)(2 + i)}{3 - i}$

1.2 Complex Plane

INTRODUCTION A complex number $z = x + iy$ is uniquely determined by an *ordered pair* of real numbers (x, y). The first and second entries of the ordered pairs correspond, in turn, with the real and imaginary parts of the complex number. For example, the ordered pair $(2, -3)$ corresponds to the complex number $z = 2 - 3i$. Conversely, $z = 2 - 3i$ determines the ordered pair $(2, -3)$. The numbers 7, i, and $-5i$ are equivalent to $(7, 0)$, $(0, 1)$, $(0, -5)$, respectively. In this manner we are able to associate a complex number $z = x + iy$ with a *point* (x, y) in a coordinate plane.

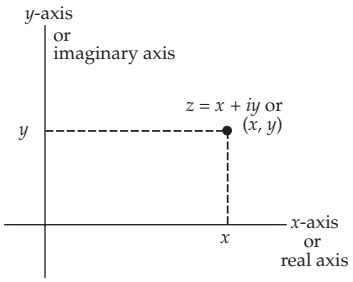


FIGURE 1.2.1 z-plane

Complex Plane Because of the correspondence between a complex number $z = x + iy$ and one and only one point (x, y) in a coordinate plane, we shall use the terms *complex number* and *point* interchangeably. The coordinate plane illustrated in FIGURE 1.2.1 is called the **complex plane** or simply the **z-plane**. The horizontal or x-axis is called the **real axis** because each point on that axis represents a real number. The vertical or y-axis is called the **imaginary axis** because a point on that axis represents a pure imaginary number.

Vectors In other courses you have undoubtedly seen that the numbers in an ordered pair of real numbers can be interpreted as the components of a vector. Thus, a complex number $z = x + iy$ can also be viewed as a two-dimensional position **vector**, that is, a vector whose initial point is the origin and whose terminal point is the point (x, y). See FIGURE 1.2.2. This vector interpretation prompts us to define the length of the vector z as the distance $\sqrt{x^2 + y^2}$ from the origin to the point (x, y). This length is given a special name.

A brief review of vectors can be found in Section 1.2 of the student study guide. ▶

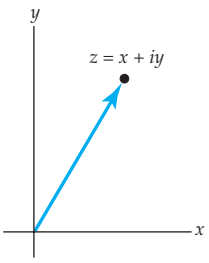


FIGURE 1.2.2 z as a position vector

DEFINITION 1.2.1 Modulus

The **modulus** of a complex number $z = x + iy$ is the real number

$$|z| = \sqrt{x^2 + y^2}. \tag{1}$$

The modulus $|z|$ of a complex number z is also called the **absolute value** of z. We shall use both words *modulus* and *absolute value* throughout this text.

EXAMPLE 1 Modulus of a Complex Number

If $z = 2 - 3i$, then from (1) we find the modulus of the number to be $|z| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$. If $z = -9i$, then (1) gives $|-9i| = \sqrt{(-9)^2} = 9$. ☰

*Throughout this text we shall use the abbreviation CAS for "computer algebra system."